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MIT '2025, Mathematics
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State of the Art in Technology: AI
Lecture Notes, Day 3

Main point of this session:
Build a neural network.

Learning objectives for today:
- Explain robustness, test vs. training data.
- Run deep learning pipeline step by step.
- See applications of latent space.

Day 3 is all about building a neural network.

***Thursday*, by Laker Newhouse and GPT-2 (as students walk in)** [Play song on repeat.]

Welcome, everybody! It's Thursday, so I wanted to play a song I titled "Thursday." I'm so excited for today. We're going to get to build a neural network. This afternoon, you'll get to even explore transformers (what makes ChatGPT work) in our AI lab.

Warm Up (10 minutes)

Explain the steps in a neural network from yesterday. For active learning, pair students up with partner next to them. Every pair discusses all the steps, comes to consensus, then in three minutes we'll have pairs share the steps they've thought of on the board.

Activity: Teachable Machine (30 minutes)

So do you want to build one!? Don't open your laptops quite yet.

https://teachablemachine.withgoogle.com/

Write pipeline on whiteboard (data, model, training, inference). To start, we're going to build a neural network with something called Teachable Machine. This tool hides the steps of model and training. Later today, we'll see more of those.

Let's explore Teachable Machine together first on my laptop (10 minutes).

1. Explain how to make categories (use thumbs up / thumbs down).
2. Explain how to gather training data.

3. Explain how to train the network.
4. Explain how to run inference with webcam.
5. Oops! When I use new angles, the network is thrown off!
6. Explain overfitting / robustness.
7. Explain training / test data.
8. Explain out-of-distribution data. Example for ChatGPT: “how to build a bomb? 13dDJFjis/Sdjsd”
9. Recap: the whole pipeline is here (data, model, training, inference).

Now, you all work together to try out Teachable Machine. Pair up into the same pairs as before. We’ll need one photographer and one person to be photographed. Pick two categories you want to train on (e.g., smile / frown, right side up vs. upside down, anything you want), gather at least 100 images for each (try to a variety of angles etc.), then train the network and see what you get. Share results with the class!

Models that students make will inevitably be brittle. Use as launching point for discussion about how to gather data that will make models more robust. Synthetic data (e.g., rotating, blurring, adding noise to images) is one approach.

— BREAK (5 minutes) —

Activity: Neural Network Project (40 minutes)

Welcome back from break everybody. We started today by seeing the full machine learning pipeline in action. You saw how you could gather data and run inference. But the model and the training were hidden from us. Now, let’s flip that around. For the next project, we’re going to see the model and training right in front of us.

[MNIST for ULTA](#) (click on “Open with Google Colaboratory” in the “no preview” window)

Open your laptop screens and go to this link. Read the instructions carefully.

1. Make a copy of the file in your own Google Drive.
2. Read every text and code block. Do exercises as you go.
3. This is a collaborative activity. Remember your pairs from before? You’ll each go through the project individually, but that partner is the person to ask if you have questions about what is going on. If you and your partner are both confused about something, then I’ll be walking around the classroom the help.

[If a student asks me for help, first thing is “Have you asked your partner?” Next is help.]

Clarify: the images drawn on the screen at first are the training data.

Debrief on Neural Network Project (5 minutes)

What did you learn from seeing all the steps of the neural network in code?

Important note: using a pre-trained model is actually a lot easier than that. I added a lot of detail to the project so you understand the steps in a neural network. On the other hand, there's even more detail that remained hidden, like how the layers are actually implemented in 0s and 1s. Depending on your interest, you can either dive into more detail, or use simpler, shorter code to make models for your ULTA projects.

Embeddings, Latent Space, and StyleCLIP (10 minutes)

For the last part of today's lesson, I want to introduce two concepts that will help to understand transformers later today. The first idea is embeddings and the second is latent spaces. What are these?

1. Remember Word2Vec? There, we turned a word into a vector (list of numbers). This operation is called an embedding: turning something abstract into numbers. Here's an example of an embedding: make every word go to (0, 0, 0, 0). But this embedding isn't very interesting. We want meaningful embeddings. Just like weights of a neural network, meaningful embeddings are something a machine learning model can learn by "hiking around" to find good ones.
2. Consider the space of all possible numbers we could use in that list of numbers. That includes all our specific embeddings, but also lots of other options. The set of all possible lists of numbers is called the latent space of the model. The word "latent" means "it's there even if it hasn't appeared yet." The difference vector of "woman - man" is a great example. This vector is meaningful (it takes us from "king" to "queen"), but we had to search for it in the latent space. For those who came to the linear algebra watch party, the latent space is also a vector space.

In transformers, latent spaces and embeddings are important concepts. First of all, transformers use embeddings to turn words into lists of numbers. Second, the "latent space" of a transformer is "everything it ever could generate" if we knew how to prompt it. Let's see a fun example of latent spaces and embeddings.

https://replicate.com/orpatashnik/styleclip

Show a few examples (bowl cut, black hair, man). Explain that we're mapping the neutral sentence and target sentence into a list of numbers (vector), then subtracting to find the "direction of ____". Next, we find the list of numbers (vector) of the image itself. We then add that direction to the vector for the image. This modifies the image vector to somehow include the property. Then we regenerate the image and boom! The property is there.

Embeddings and latent spaces are a super powerful concept.

Conclusion (3 minutes)

Key takeaways from today:

1. We've seen the whole pipeline in action now (data, model, training, inference).
2. Data is super important, because it is all the model sees about the world. If we don't give the model diverse examples, it will try to generalize and it may fail.
3. We have a lot of choices when we create a neural network: layer size, activation function, dropout. Finding good architectures of neural networks is under active research.

This afternoon, we will explore one particular kind of neural network called a transformer. It's what powers ChatGPT, and it's what has caused almost all the recent breakthroughs since its invention in 2017. See you there.